**Algorithm 2** Hierarchical Pruning

1: **procedure** HQ-PRUNING($Q, S_d, \beta, EnsSet$)
2: **Input**: $Q$: the focal diversity metric; $S_d$: the desired ensemble size; $\beta$: the percentage of the number of ensemble teams to be pruned out in each iteration; $EnsSet$: the set of sub-ensemble teams to be considered;
3: **Output**: $GEnsSet(Q, S_d)$: the set of good ensemble teams of size $S_d$ identified by the focal diversity metric $Q$.
4: Initialize $pruneSet = \{\}$ ▷ Subsets of member models to prune out.
5: **for** $S = 2$ to $S_d$ **do**
6: Initialize $GEnsSet(Q, S) = \{\}$, $D = \{\}$
7: Construct $EnsSet(S)$ of size $S$ ensembles
8: **for** $i = 1$ to $|EnsSet(S)|$ **do**
9: **if** $T_i$ contains any subset in $pruneSet$ **then**
10: continue ▷ Prune out this sub-ensemble $T_i$ and avoid expanding along this branch.
11: **else**
12: $q_i = FQ(T_i)$ ▷ $FQ(T_i)$ is obtained through Algorithm 1 by using the focal diversity $Q$.
13: $D$.append($q_i$)
14: $GEnsSet(Q, S)$.add($T_i$)
15: **end if**
16: **end for**
17: $n = \beta \times |GEnsSet(Q, S)|$
18: Sort $T_i \in GEnsSet(Q, S)$ by $q_i \in D$
19: Remove $n$ sub-ensembles of the lowest ensemble diversity from $GEnsSet(Q, S)$ and add them into $pruneSet$
20: **end for**
21: **return** $GEnsSet(Q, S_d)$
22: **end procedure**

### 3.2 Hierarchical Pruning Overview and Algorithm

Our focal diversity metrics reveal some anti-monotonicity property: a superset of a low diversity ensemble has also low diversity. Specifically, a low focal diversity score for an ensemble team, e.g., $F_0F_2$, often implies insufficient ensemble diversity, that is a high correlation of its member models in making similar prediction errors, making the member models highly redundant for a large superset ensemble team, such as $F_0F_1F_2$. Hence, those large ensembles that contain a small sub-ensemble with low diversity (e.g., $F_0F_2$), such as $F_0F_1F_2$, $F_0F_2F_3$, $F_0F_1F_2F_3$, and $F_0F_1F_2F_4$ tend to have lower ensemble diversity than other ensembles with the same size and more diverse member models. Therefore, when we identify a sub-ensemble with low ensemble diversity, these large ensembles that are the supersets of this sub-ensemble can be preemptively pruned out, which motivates our hierarchical ensemble pruning approach. The pseudo code is given in Algorithm 2. Overall, our hierarchical pruning is an iterative process of composing and selecting deep ensembles for a desired ensemble team size $S_d$. *First,* we start the hierarchical pruning process with the ensembles of size $S = 2$, that is $|EnsSet(S = 2)| = \binom{M}{2} = 10(10 - 1)/2 = 45$ candidate ensemble teams. *Second,* for a given focal diversity metric, we rank candidate ensembles of size $S$, such as $S = 2$, by their focal diversity scores ($q_i$), prune out the bottom $\beta$ percentage of ensembles with low focal diversity scores, and add them into *pruneSet* as the pruning targets (Line 17~19). Here, a dynamic $\beta$ can be configured to accommodate the concrete number of candidate ensembles and diversity score distribution. A conservative strategy is recommended to set a small $\beta$, e.g., by default $\beta = 10\%$. *Third,* we preemptively prune out all subsequent ensemble teams with a larger size $S + 1$ that contain one or more pruned sub-ensembles in *pruneSet* (Line 7~16). By iterating through these

pruning steps, our hierarchical pruning can efficiently identify high quality sub-ensembles of the desired team size $S_d$ and add them into $GEnsSet(Q, S_d)$.

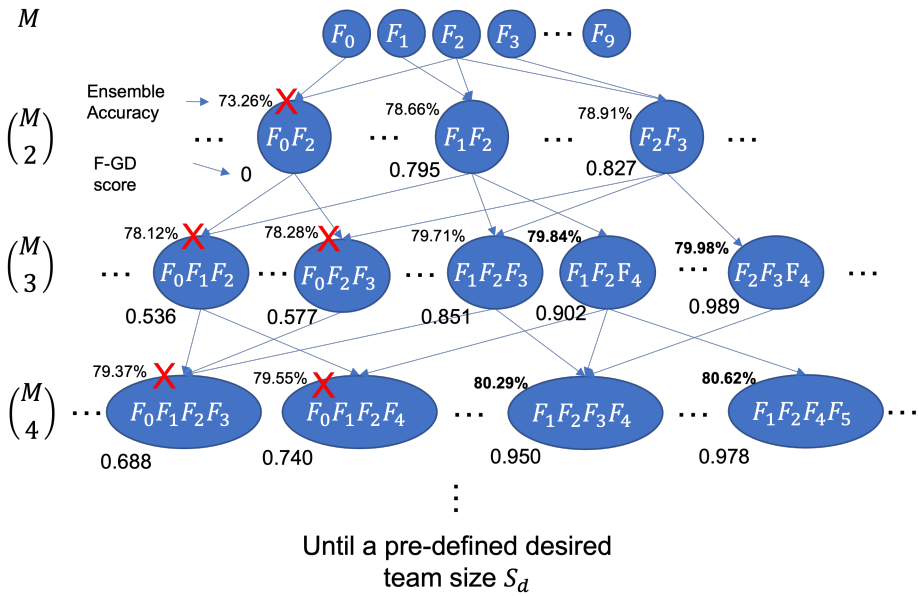
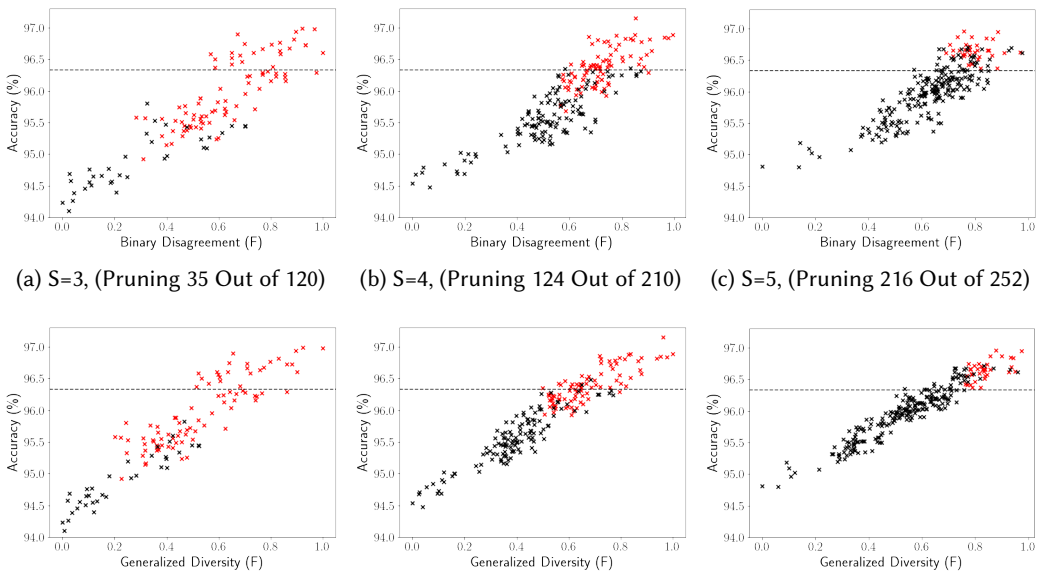



Fig. 2. Hierarchical Pruning On ImageNet

We present a hierarchical ensemble pruning example on ImageNet in Figure 2. With $M = 10$, we have all 10 member networks on the top, followed by all sub-ensembles of size $S = 2$. For each tier, we add one additional network to the candidate ensemble teams of size $S$ ($2 \leq S < S_d$) and place these extended ensembles of size $S + 1$ in the next tier until $S = S_d$, where the ensembles of the desired team size $S_d$ will be placed in the last tier. Meanwhile, for each pruned ensemble, such as $F_0F_2$, our hierarchical pruning algorithm will preemptively cut off these branches of candidate ensemble teams that contain this pruned ensemble, i.e., all ensembles that are supersets of $F_0F_2$, marked by the red cross in Figure 2. This way of building ensemble teams allows us to compose high quality sub-ensembles and strategically remove the ensembles with low diversity. As the example in Figure 2 shows that our hierarchical pruning indeed can effectively prune out these ensemble teams with low diversity and low accuracy, such as $F_0F_1F_2$, $F_0F_2F_3$, $F_0F_1F_2F_3$ and $F_0F_1F_2F_4$, and avoid exploring those unpromising branches.

We first apply our F-BD and F-GD hierarchical ensemble pruning algorithms on CIFAR-10 to prune the entire deep ensemble of 10 member networks by setting $\beta = 10\%$ and $S_d = 5$. Figure 3 visualizes the ensemble pruning results, where the black and red dots denote the pruned and selected ensemble teams respectively. Three interesting observations should be highlighted. *First,* for our focal diversity metrics, including F-BD and F-GD, the ensemble teams with high focal diversity scores tend to have high ensemble accuracy, especially when we compare the ensemble teams of the same size $S$, e.g., $S$=3, 4, 5, which is an important property to guide ensemble pruning. *Second,* our hierarchical ensemble pruning approach can effectively identify these ensemble teams with insufficient ensemble diversity by their low focal diversity scores. The effectiveness of our hierarchical pruning can be attributed to two factors. On the one hand, our focal diversity powered hierarchical pruning encourages more fair comparison of focal diversity scores among these

ensembles of the same size $S$. Comparing to Figure 1b showing all possible ensembles of mixed team sizes, Figure 3 shows much clearer correlation between the focal diversity score and ensemble accuracy for the ensembles of the same size $S = 3,\ 4,\ 5$. On the other hand, our focal diversity metrics can more precisely capture the failure independence of member networks of a deep ensemble with some anti-monotonicity property, ensuring that the focal diversity based ensemble pruning is highly accurate and efficient. *Third,* the time and space complexity of the hierarchical pruning algorithm and pruned ensemble execution cost can be ultimately bounded by the desired team size $S_d$. For example, we recommend setting the desired ensemble team size $S_d$ up to $50\% \times M$, which allows our hierarchical pruning to effectively find these sub-ensembles that offer on par or improved ensemble accuracy over the entire ensemble of $M$ member networks and have a substantially smaller team size, such as one third or one half of $M$ with a significant cost reduction in ensemble execution. *Finally,* for Figure 3c and 3f, all the selected sub-ensembles (red dots) with the desired team size $S_d = 5$ provide higher ensemble accuracy than the reference accuracy of 96.33% for CIFAR-10, which leads to 100% precision of our hierarchical pruning with F-BD and F-GD. We observe similar ensemble pruning results using the other two focal diversity metrics, F-CK and F-KW.




(a) S=3, (Pruning 35 Out of 120) (b) S=4, (Pruning 124 Out of 210) (c) S=5, (Pruning 216 Out of 252)

(d) S=3, (Pruning 35 Out of 120) (e) S=4, (Pruning 124 Out of 210) (f) S=5, (Pruning 215 Out of 252)

Fig. 3. Deep Ensembles of Size $S = 3, 4, 5$ on CIFAR-10: top three figures for **F-BD** and bottom three figures for **F-GD** ($\beta = 10\%, S_d = 5$)

**Focal Diversity Pruning by Focal Diversity Consensus Voting.** Using different focal diversity metrics in the hierarchical ensemble pruning may produce different sets of selected ensembles (*GEnsSet*). It is observed that these ensemble teams that are chosen by the majority of our focal diversity metrics in ensemble pruning (e.g., F-BD pruning, F-KW pruning, F-GD pruning, etc.) tend to consistently outperform the entire deep ensemble in terms of the ensemble accuracy and execution cost. Therefore, we introduce the third step in our hierarchical ensemble pruning, which consolidates these ensembles that are selected by different focal diversity metrics into a majority voting based final selection. Concretely, an ensemble team will only be added to this final selection